

Apple violates laws

Apple under scanner for violating competition laws in India for restricting sale of iPhone to only two operators

Violent games reduce crime

According to a paper by Centre for European Economy research and University of Texas, violent video games are a factor in reducing crimes in US

Mobile computing

Best of both worlds?

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Consider this – you have a few important emails to send and the battery life indicator on your notebook is in the red. “15 minutes remaining”, it warns nonchalantly while you hit panic mode with no plug point in sight. Try whatever power management tweaks you may, but realistically you know in your heart that the notebook won’t be with you for more than ten minutes. What do you do? Nothing really, other than gnaw at your fingernails till you’re left with stubs for fingers.

Unless of course your notebook was a hybrid computing device, like one that’s based on Cupp Computing’s PunkThis module. In this case, you could easily flip a switch and instantly start working on an ARM-based OS, like a modified version of Android 2.3 and make the same battery last for eight hours. How? We’ll get to that in a bit, but first a background for those who came in late.

The story so far...

The two operational scenarios we described above (PC on x86 and Android on ARM) are the result of two distinct underlying platforms that have been at war for sometime. Thus far both factions of this brewing war have had distinctly marked out territories with x86 dominating the performance market, and ARM resulting in shipping of unbelievably large quantities of chips for embedded systems and portable devices.

The x86 architecture which is used by Intel’s processors was designed with some specific goals in mind – performance and manufacturing scalability. Microprocessors from Intel that are descendants of the 8086 chip developed in the late seventies are all part of the x86 family. The architecture was quickly adopted and was responsible for the demise of many processors of that era. Even Apple toed the line

While the impending ARM v/s x86 battle looms on the horizon, there is perhaps some middle ground being eked out; a truce of sorts. Read on to find out how we may be able to have best of both worlds soon and how the mobile computing landscape may change.



The one that sired them all

eventually by dropping IBM’s PowerPC architecture in favour of the strides Intel was making year on year. This is what makes Intel dominant in the PC and server space, but not in the mobile space because of the huge power consumption demands of the chips.

ARM’s architecture, on the other hand, was focused on low power consumption right from the beginning. Chips based on ARM’s IP and version seven instruction set are brought to market by various chipmakers including Qualcomm, Texas Instruments, NVIDIA etc, while for x86, Intel has competition from AMD. As for the ARM architecture’s shortcomings? It doesn’t do so well with multi-tasking and high performance tasks.

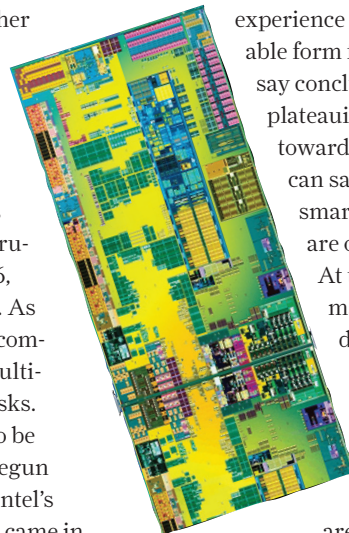
PCs, initially, weren’t meant to be portable and have only later on begun to focus on power management. Intel’s biggest stride in the mobile space came in the form of its Oak Trail platform released

recently. Last month at Computex we saw Oak Trail tablets from the likes of ViewSonic which have better performance in terms of power consumption – about eight hours – though not nearly as much

as ARM-based tablet devices.

So what do we have now? A whole bunch of generic names for devices – MIDs, ultraportables, Pocket PCs, Smartphones – and the lines between them are constantly blurring, as their functions converge. The ultimate goal is to

give you as complete a computing experience as possible in a portable form factor. While we can’t say conclusively that PCs are plateauing or even heading towards a decline, what we can say with certainty is smartphones and tablets are on an upward swing. At the same time, no matter how converged devices become, people aren’t carrying that one ‘all powerful’ device yet. Function-specific devices are still what works for most people. Can you



Oak trail die